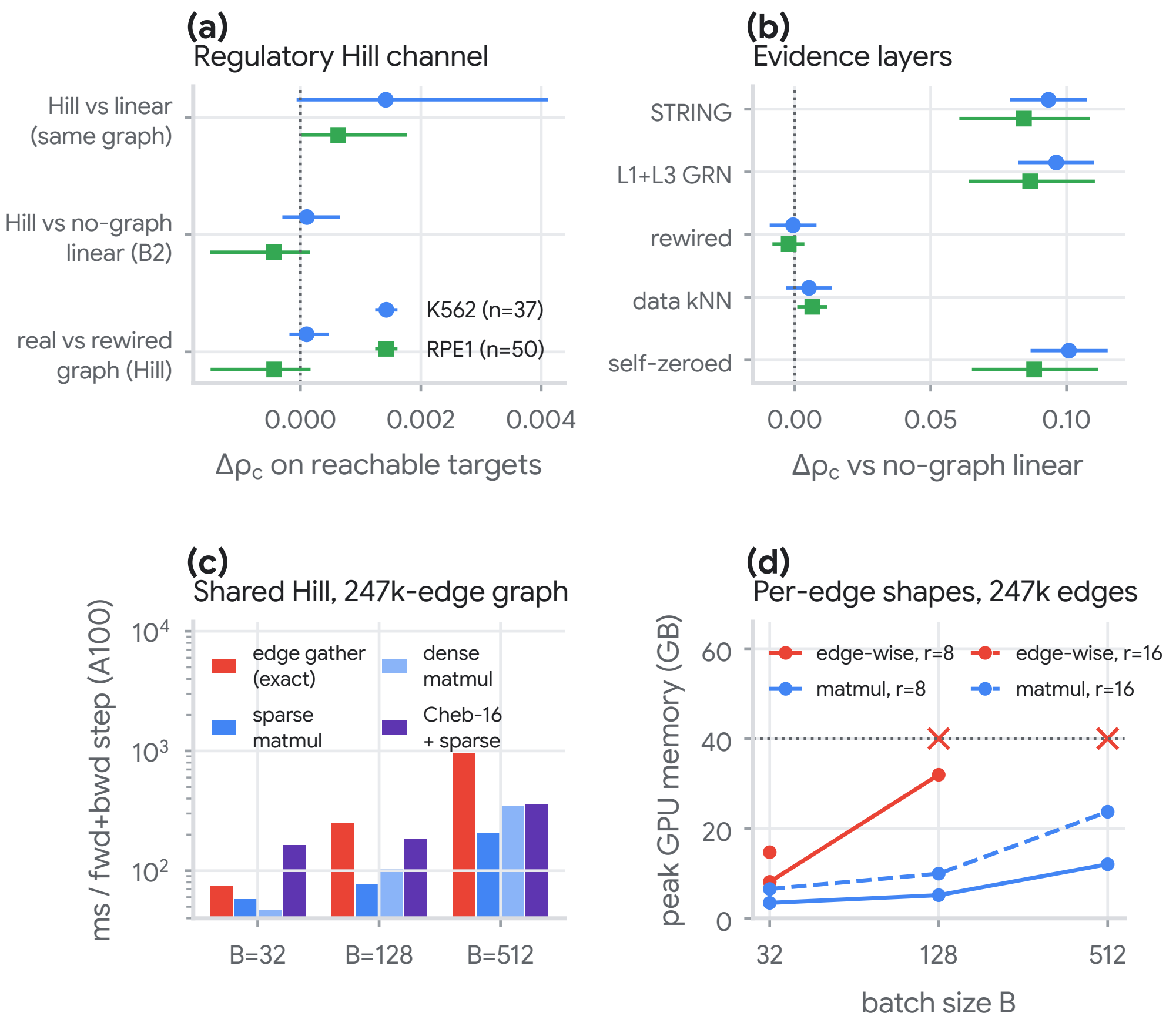
(a)
Regulatory Hill channel
Hill vs linear (same graph)
Hill vs no-graph linear (B2)
real vs rewired graph (Hill)
K562 (n=37)
RPE1 (n=50)
0.000
0.002
0.004
Δρc on reachable targets
(b)
Evidence layers
STRING
L1+L3 GRN
rewired
data kNN
self-zeroed
0.00
0.05
0.10
Δρc vs no-graph linear
(c)
Shared Hill, 247k-edge graph
ms / fwd+bwd step (A100)
10^4
10^3
10^2
edge gather (exact)
dense matmul
sparse matmul
Cheb-16 + sparse
B=32
B=128
B=512
(d)
Per-edge shapes, 247k edges
peak GPU memory (GB)
60
40
20
0
edge-wise, r=8
edge-wise, r=16
matmul, r=8
matmul, r=16
32
128
512
batch size B